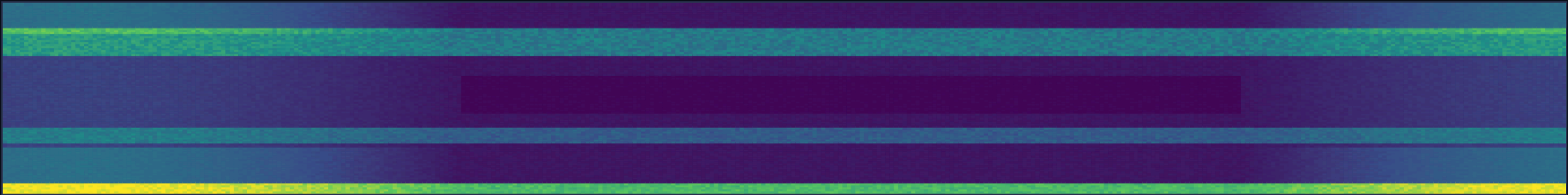


Energy fingerprints

What a 15-minute meter reading tells you about a home — and, just as often, what it does not.



00:00 → 24:00 · one household · 365 days

kWh / 15 min

82,476 METERS · 1,981 SURVEYED HOMES · 5 ASSETS · 1 EVALUATION FRAMEWORK

Identify what a household owns, from the meter alone

No visit, no survey, no questionnaire. Just import and export every 15 minutes.

PV

GENERATION

does it produce?

Battery

STORAGE

does it store?

Heat

HEAT PUMP

how does it heat?

EV

CHARGING

is a car plugged in?

Boiler

HOT WATER

resistance or pump?

| AEW — 82,476 metering points, 15-minute import and export, January 2023 to July 2026

| CKW — 1,981 surveyed households, 20 months of net metering, plus MeteoSwiss weather

| Graded on accuracy, robustness, explainability and innovation

| The hard part was never the detection. It was knowing whether we were right.

1,192 register entries. 208 usable labels.

Where supervised ground truth quietly disappears.

1192 register entries

878 distinct customers

289 rows are repeat installations by the same household

337 linked to a meter

541 customers have no meter anywhere in the delivered mapping

208 usable

after short histories and multi-meter customers

62% of subsidised customers have no meter link. Same six-digit format, same numeric range — they are simply absent from the file.

And the labels that survive are not ground truth

The register records **subsidies**, not assets. A dash means "no subsidy record", not "no asset".

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FLAGGED "NO PV"

...were exporting to the grid. The negatives are not negatives.

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PV METERS TESTED

...were already exporting *before* their recorded commissioning date, by 31 to 457 days.

0

CLEAN TRANSITIONS

Not one meter switched cleanly from silent to producing on the registered date.

So precision is **not computable** for heat pumps or EVs on AEW data — only recall, and a flag rate over the unlabelled pool

For PV we abandoned the register entirely: measured export is objective, covers every meter, and dates commissioning to within days

This is a finding for AEW in its own right — their register understates asset ownership in their own network.

Two datasets, complementary weaknesses

Develop where you can measure. Apply where the question actually lives.

AEW

82,476 metering points · 3.5 years

Import and export measured separately

208 usable labels, from a subsidy register

Negatives unusable — a dash means no paperwork

CKW

1,981 households · 20 months

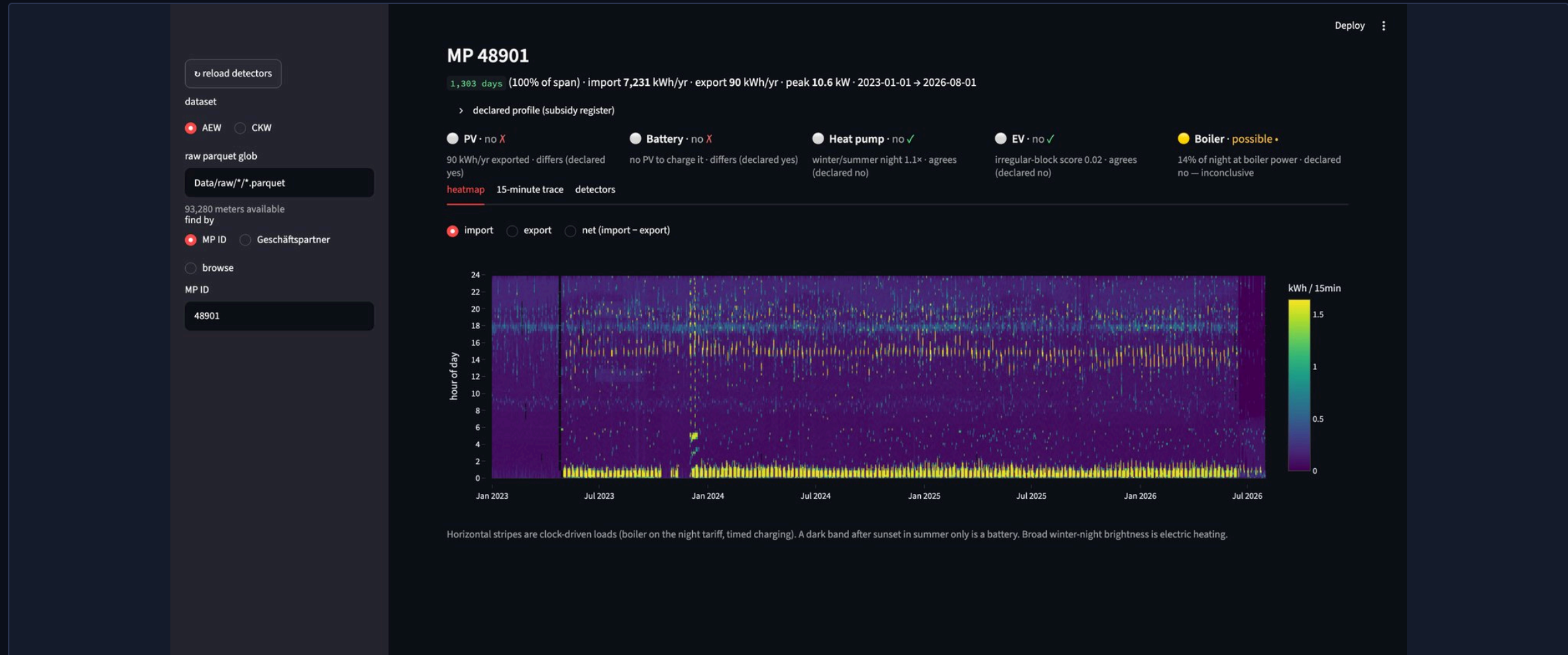
Net metering, plus hourly MeteoSwiss temperature

1,981 usable labels, from a household survey

Negatives are real — someone answered "no"

Build and validate on CKW, where a "no" means no. Then run the detectors across AEW's 82,476 meters, where the challenge lives.

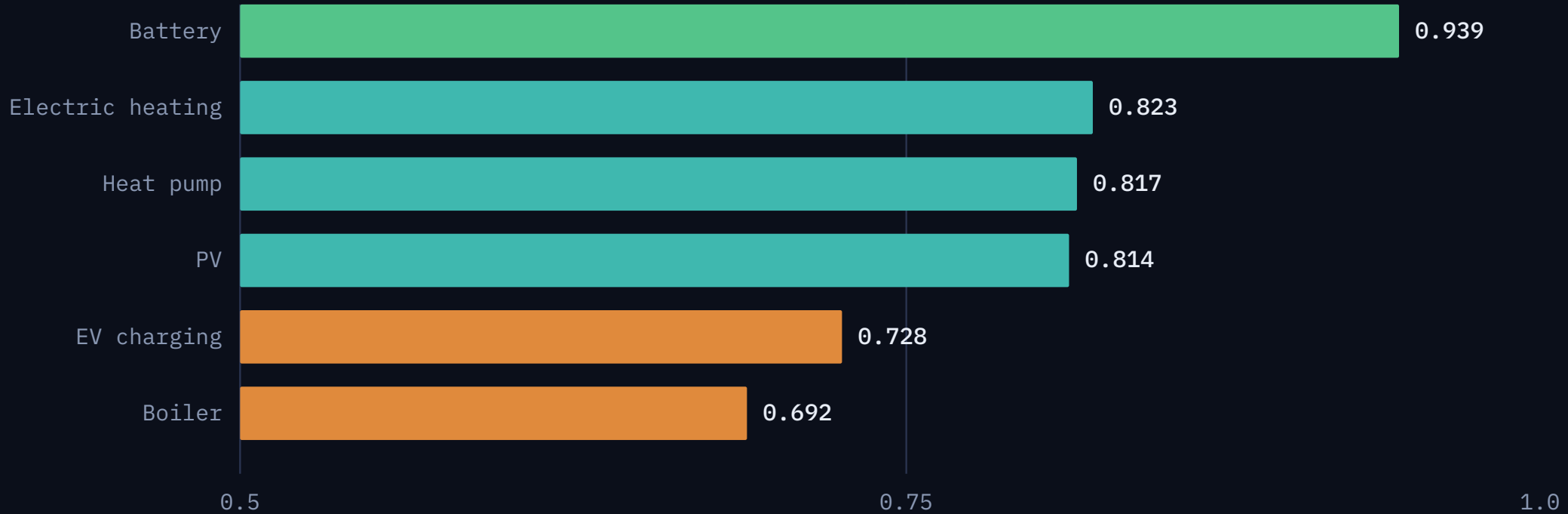
Every meter, inspectable in one screen



Hour of day against date, 1,303 days. **Horizontal stripes are clock-driven loads** – this meter fires a 5.7 kW block at midnight and a 1.9 kW block at 16:30, every day, for three and a half years. Verdicts sit above, each compared against the declared label.

What we can actually detect

Out-of-fold AUC with 95% bootstrap intervals. Random baselines: 0.49 to 0.55.



A battery is the easiest thing in the dataset: net import sits flat at **zero** after dark — 92% of dark summer hours for owners, **0%** for everyone else.

Physics or machine learning?

Paired bootstrap on the same meters — far more sensitive than comparing two intervals.

ASSET	PHYSICAL RULE	GRADIENT BOOSTING	VERDICT
PV	0.814	0.797	not distinguishable
Heat pump	0.798	0.817	not distinguishable
Electric heating	0.802	0.823	not distinguishable
EV charging	0.714	0.728	not distinguishable
Battery	0.939	0.840	rule wins
Boiler	0.606	0.692	model wins

Where a mechanism exists, one feature derived from it beats a dozen weak ones. Where none exists — resistance water heating — only a model gets anywhere.

Each detector is a physical statement

Not a fitted curve. That is what makes them explainable, and what makes them transfer.

ASSET	AUC	WHAT IT KEYS ON
Battery	0.939	Net import flat at <i>zero</i> through dark summer hours — a battery charged by day carries the evening.
Heat pump	0.817	Overnight consumption regressed on heating degree hours. The slope is heat loss ÷ COP — which also separates a heat pump (3.5×) from resistance heating (10.2×).
PV	0.814	Measured export. Sustained export <i>is</i> generation — no register needed, and it dates commissioning to within days.
EV	0.728	Wallbox-band blocks weighted by start-time <i>irregularity</i> . A car is plugged in when the driver gets home; a timer fires at the same minute.

A timer is not a car

↕ reload detectors

dataset

● AEW

○ CKW

raw parquet glob

Data/raw/*/*.parquet

93,280 meters available
find by

● MP ID

○ Geschäftspartner

○ browse

MP ID

48901

Deploy

MP 48901

1,303 days (100% of span) · import 7,231 kWh/yr · export 90 kWh/yr · peak 10.6 kW · 2023-01-01 → 2026-08-01

> declared profile (subsidy register)

● PV · no X

90 kWh/yr exported · differs (declared yes)

● Battery · no X

no PV to charge it · differs (declared yes)

● Heat pump · no ✓

winter/summer night 1.1× · agrees (declared no)

● EV · no ✓

irregular-block score 0.02 · agrees (declared no)

● Boiler · possible •

14% of night at boiler power · declared no — inconclusive

heatmap15-minute tracedetectors

asset	detector	score	confidence	evidence
battery	battery_dark_zero	0.1557	high	summer_dark_zero_frac=0.156
battery	battery_seasonal_gap	0.1557	high	summer_zero_frac=0.156, winter_zero_frac=0.0
boiler	boiler_night_band	0.1395	low	night_frac_over_1_8kw=0.1395
boiler	boiler_tariff_contrast	2.1301	low	tariff_window_kwh=0.414, afternoon_kwh=0.1943
electric_heating	eh_temp_slope	None	moderate	no weather joined - pass --weather
ev	ev_peak_band	0.9611	IGNORE — worse than chance	peak_kw=10.6
ev	ev_sustained_blocks	0.0167	low-moderate	blocks=1123, days=1299, blocks_per_day=0.865
hp	hp_night_share	-0.0415	moderate	winter_night_share=0.336, summer_night_share=0.378
hp	hp_winter_ratio	1.0732	moderate	winter_night_kwh=8.82, summer_night_kwh=8.22
hp	hp_temp_slope	None	moderate	no weather joined - pass --weather

Confidence is each detector's measured AUC on labelled data — battery 0.94, PV ~0.90, heat pump 0.83, EV 0.74, boiler 0.64 — not a statement about this meter. A blank score means the detector could not apply (too little history, or no weather joined).

Every score carries the evidence that produced it. **ev_sustained_blocks: 0.0167** — 1,123 blocks across 1,299 days, but they start at the same minute, so the detector rejects them. **ev_peak_band is flagged IGNORE:** measured AUC 0.349, reliably worse than chance, kept on the board as a warning rather than deleted.

The framework found it, not us

0.78

BEFORE

A 5.7 kW block on 78% of days. Counting blocks, it looked like someone charging a car every evening.

0.02

AFTER

Weighted by start-time irregularity, the same meter is correctly rejected. And the rule then *matched* the gradient-boosted model.

Swiss 300 L Elektroboiler ship at **3.0, 4.0 and 6.0 kW** — overlapping the wallbox band (3.4–12.8 kW) almost entirely

Power cannot separate them. **Timing can.**

Six detector ideas were tried and measured. Three survived — the three that did not are documented with their numbers, so nobody repeats them

This is the argument for building the evaluation harness before the detectors, not after.

Four things that would mislead you

All measured on the delivered data. None hypothetical.

ASSETS MASK EACH OTHER

A battery holds net import at zero after dark — erasing the evidence a boiler detector reads. The same rule implies 48.5% boiler prevalence without a battery and 29.8% with one. Detect batteries first.

THE EVAL SET IS NOT THE FLEET

AEW's labelled meters are 84% PV and 64% battery; CKW's respondents 72% solar and 77% heat pump. Both wildly over-represent owners. Never quote a prevalence from them.

NEITHER STORES LOCAL TIME

AEW is fixed CET, CKW is UTC. A ripple-controlled load therefore appears at two different hours across the year, smearing every time-of-day feature. One meter's start-time concentration moved from 0.72 to 0.88 once corrected.

DST IS LOSSY AT SOURCE

The export writes a fixed 96-slot grid, so the October hour is simply absent and the March day is padded. No amount of processing recovers it.

What would move this furthest

The bottleneck is labels, not method.

1 The missing partner-to-meter links

541 subsidised customers have no meter in the mapping. Recovering them multiplies the labelled set by 2.6 — the largest single gain available, and it needs no modelling work at all.

2 Weather joined by postcode

The strongest heat-pump detector regresses consumption on heating degree hours. AEW's load file already carries the postcode; it simply has no temperature to join to.

3 A survey, even a small one

CKW's 1,981 declared households are the only reason heat pumps and EVs are measurable at all. A few hundred AEW customers would do the same for their own network.